

Early Diagnosis of Autism Spectrum Disorder in Children Using Machine Learning

Dr.Mrs.S.Sasikala
Head of the Department,
Department of Data Analytics (PG).
PSGR Krishnammal College for
Women,Coimbatore.

Dencil Jaushmy J
Department of Data Analytics (PG).
PSGR Krishnammal College for
Women,Coimbatore.

ABSTRACT

The enhancement in neurodevelopmental disorders, especially in Autism Spectrum Disorder (ASD) detection and intervention, determines the developmental fate of an individual. ASD is a heterogeneous clinical diagnosis that interferes with communication, social behavior, and interaction; within this context, the presentation of ASD is more varied than in any other disorder, especially among children. Early identification and individualized support can have a positive impact on a child's learning, social skills, and overall quality of life. The present study moves on to investigate an integrated multimodal dataset of facial image stimuli, eye-tracking scanpath data, and metadata, including parameters such as age, gender, ASD classification, and clinical severity gauged by the Childhood Autism Rating Scale (CARS). The eye-tracking data, which was recorded as sequential gaze coordinates in JSON files, were overlaid on facial images to visualize the fixation and saccade movements. The dataset divides into ASD and Non-ASD groups and was then used for characteristic comparisons of gaze behaviors. Preprocessing involved exercising gaze data acquired, rectifying gaze trails through extracted temporal stimuli, and feature extraction. Data from gaze features and their correlation with CARS score were classified and the severity of ASD predicted through Random Forest, Logistic regression, and Support Vector Machine (SVM). Eye-gaze strategies distinguished considerably between the groups of ASD versus Non-ASD children, with ASD subjects presenting with lesser facial fixations and disorders with it-irregularities in their scanpaths. These findings highlight the added value of combining eye tracking with interpretable machine learning for supporting the ASD screening process. The research offers insights of considerable clinical relevance for professionals and educators, allowing them to contribute toward the emergence of personalized diagnostic and intervention tools. It also portrays the impact of data science within pediatric neurodevelopmental assessment and early detection of ASD.

Keywords-Machine Learning,, Logistic Regression, Support Vector Machine (SVM), Autism Spectrum Disorder (ASD),Random Forest, Eye-Tracking ASD.

1. INTRODUCTION

Global prevalence of Autism Spectrum Disorder has kept rising steadily over the decade, leading to increased focus on early identification and intervention. The understanding of neurodevelopmental disorders is quickly growing, and so is the need for objective scalable tools in their diagnosis. Most traditional screening methods rely heavily on subjective clinical assessment, delaying diagnosis and support. Emerging technologies such as eye tracking are opportunities for observing behavioral markers in a non-invasive and quantifiable manner. Coupled with machine learning, these technologies could completely change how ASD is screened and interpreted and possibly managed.

1.1 Understanding Autism Spectrum Disorder and Diagnostic Challenges

ASD is a complex neurodevelopmental disorder that manifests itself through enduring differences in social communication, interaction, and behavior. Although symptoms usually start during early childhood, they differ from one individual to another, making ASD highly heterogeneous and difficult to diagnose. Early diagnosis is essential, and it allows for timely interventions that result in major improvements in developmental outcomes. On the contrary, traditional general methods of diagnosis-the clinical observations and behavioral assessments-are usually long, subjective, and unavailable in many places. Future dependency will be compared with scalable, objective technology-assisted screening solutions.

1.2 Eye-Tracking and Machine Learning for Early ASD Detection

Eye-tracking techniques present a powerful and non-invasive method for studying visual attention and social perception, which may be skewed in various ways in individuals with ASD. By tracking how children view facial stimuli, the researchers are able to distinguish between unique look patterns that may reflect neurodevelopmental differences. Our project extracts statistical image features, including mean intensity, standard deviation, and active pixel count, to quantify these visual behaviors in eye-tracking scan path images. The eye-tracking features are joined with other information, such as age, gender, and Childhood Autism Rating Scale (CARS) scores, for training machine learning models. The classifiers Random Forest, Support Vector Machine (SVM), and Logistic Regression are being used in our research to predict the presence and levels of severity of ASD.

1.3 Study Objective and Significance

The main goal of this study is the development of a lightweight, interpretable, reproducible classification pipeline for detecting ASD traits and estimating severity from image-derived features. The feasibility of early autism detection using simple scanpath statistics and demographic metadata, without the need for highly specialized hardware or complex imaging, is emphasized. This development could greatly enhance clinical diagnostic support while also fostering the emergence of scalable digital screening tools, particularly in under-resourced settings. Machine learning combined with eye-tracking data thus

further the efforts toward creating early, accessible, and data-driven autism diagnostics.

2. LITERATURE REVIEW

Jiang et al. (2019) used Random Forest classifiers applied to task-related gaze and behavioral data. The model achieved 91% sensitivity in identifying discriminative multimodal feature sets and illustrated the advantages of combining visual and task-level cues. Liu et al. (2016) analyzed the face-scanning patterns and trained SVMs on the corresponding eye movement data. They classified subjects with 88.5% accuracy as either ASD or TD children. Particularly useful was looking at the regions of the eyes and mouth. Tsuchiya et al., (2021) developed application of Gazefinder for recording visual attention in school-aged children. They applied the best-fit age-based model to yield classification accuracies of approximately 78%. It has shown that age-sensitive models are feasible for real-field screening. Xie et al. (2019) have developed a deep neural network that analyzes eye-tracking scanpath images for classifying autism spectrum disorder (ASD). The deep learning model performed about 95% accuracy on test stimuli with salient pixel-based features. The results showed that convolutional architectures are useful in scanpath analysis. Wan et al. (2019) employed Support Vector Machines (SVM) to categorize ASD and TD groups using fixation durations. More than 80% accuracy, sensitivity, and specificity were attained. This work stemmed from a motivational argument emphasizing a temporal gaze metric toward ASD diagnosis. Elbattah et al.

(2019) clustered scanpaths into low-dimensional representation by unsupervised autoencoders without predefined labels or groups. The formed clusters nearly corresponded to groups categorized under ASD and TD. The effectiveness of unsupervised learning on visual behavior was shown through this approach. Cilia et al. (2021) employed CNNs to process consecutive eye movement images for prediction purposes in patients diagnosed with ASD. On average, their model classified individuals with 90% confidence. It explained the potential of deep learning to exploit fine-grained differences in gaze pattern. Kanhirakadavath & Chandran (2022) constructed motion based gaze features for deep neural networks. They reported an increase in model accuracy by 23% using velocity and jerk metrics. This study is based on dynamic eye behavior and not static fixations. Lin et al. (2022) combined task-based gaze patterns and task metadata in tree-based classifiers. Random Forest had the highest performance at 70% accuracy in autistic trait prediction. It demonstrated the strength of hybrid gaze-demographic models. Kong et al. (2023) incorporated SVM on toddler gaze data with demographic features. It classified early ASD detection with ~80 percent accuracy. Concentrating on simple ML pipelines provided the feasibility for early intervention. aneva and others (2018) were eye-tracking study for use during web interactions in adolescents with ASD. More or less accurate up to 75% based upon logistic regression for class separation. This was proved that gaze can reflect cognitive patterns beyond the stimuli face. Akter et al. (2023) benchmarked ML classifiers on

scanpath images using MLP, SVM, and XGBoost. MLP achieved 87% accuracy post-cluster-based preprocessing. It demonstrated how pre-clustering can strengthen classifier robustness. Zhao et al. (2024) analyzed gaze metrics and also built ensemble models to detect ASD from a large dataset. Random Forest attained $AUC \approx 0.85$ and emphasized non-social gaze as a key predictor. The contribution of this study was strong generalization with the clinical-scale data. Rashid & Shaker (2024) evaluated ML and deep learning models using public eye-tracking datasets. Competitive with SVM and RF with respect to small-sample conditions against CNNs. Supporting evidence that under such conditions, ML has viability in low-resource settings. Mohammadifar et al. (2023) created a federated SVM framework for multi-center prediction of ASD. Data privacy was ensured while still maintaining more than 85% classification accuracy. It showcased secure and collaborative learning in medical ML.

3. EXISTING METHODOLOGY

In recent years, early detection of Autism Spectrum Disorder (ASD) has become a passion in interdisciplinary research, gathering components from areas such as clinical psychology, neuroscience, and artificial intelligence. Traditional methods for diagnosing ASD, such as behavioral assessments and parent questionnaires, although accurate, suffer from their subjectivity, time-consuming nature, and resultant delay in diagnosis. To overcome this, researchers have looked at applying an eye tracker to machine learning techniques to

objectively establish visual attention patterns specific to ASD.

Several studies have suggested that gaze behaviors of children with ASD tend to differ, such as less fixation on facial features and social stimuli. An eye tracker can record these types of gaze behaviors very finely, generating variables such as fixation duration, saccade frequency, and scanpath heatmaps. These parameters are fed into classifiers that use supervised machine learning such as Support Vector Machines (SVM), Random Forest, Logistic Regression, K-Nearest Neighbors (KNN), and Convolution Neural Networks (CNNs). The goal of these methods is to discriminate between gaze behavior from children with ASD and those who are neurotypically developing.

In their existing form, models have done well in terms of classification accuracy, sensitivity, and generalizability. In that regard, Random Forests have shown resilience in the presence of noisy datasets, while the SVM work best if the data distribution is small and well separated. CNNs and RNNs for deep learning have been employed for gaze sequence data and scanpath images for automatic feature extraction.

Even though some progress has been made, the majority of the existing systems have major deficits including small sample size, lack of diversity in the training dataset, and lack of clear explainability of the model outputs. Also, in many instances, the models relied only on gaze features without much incorporation of other complementary metadata such as age, gender, or CARS

scores, which lay candidates for increasing diagnostic precision and giving insights about estimating the severity of ASD. This essentially presents a gap towards the generation of a more comprehensive, multimodal, and explainable framework for ASD prediction.

4. PROPOSED WORKFLOW FOR ASD DIAGNOSIS

This study aims to develop efficient machine learning pipeline to classify children with Autism Spectrum Disorder (ASD) traits as compared with Non-ASD children based on eye-tracking scanpath images. The approach consists of ways to carry out image preprocessing, statistical feature extraction, supervised learning algorithms, and model evaluation, thus making it very helpful for the early diagnosis of the conditions.

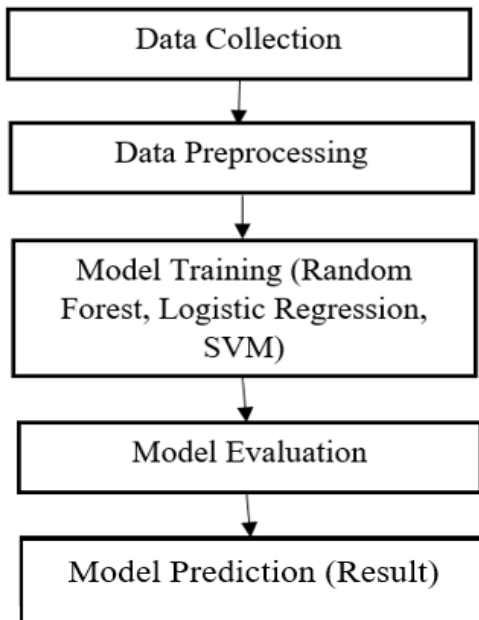


Fig 4.1 Workflow of the Proposed System

This diagram illustrates the essential stages of an ASD detection pipeline using eye movement images. It comprises data

collection and preprocessing to derive pixel-level statistical features. Then, these features are used to train machine learning models such as Random Forest, Logistic Regression, and SVM. The models are evaluated for accuracy and robustness, using standard classification metrics. Finally, predictions are made on such images to ascertain whether a child is likely ASD or not.

4.1 Data Preprocessing

4.1.1 Data Cleaning and Transformation

The dataset initially encompasses image metadata and the corresponding scanpath features having missing values or inconsistent data types. The preprocessing started by handling null values; those were imputed or dropped depending on their significance to the dataset. Numerical features such as mean intensity, standard deviation, and active pixel count were created from grayscale images based upon custom image processing functions. These were then transformed to structured formats that are compatible with machine learning workflows. The labels of ASD and Non-ASD were encoded in binary classes as TS=1 and TC=0, respectively, for supervised classification tasks.

4.1.2 Encoding and Feature Engineering

The categorical attributes Gender and diagnosis classes were label encoded to save semantic meaning but conform to model interoperability. The image paths were built programmatically for batch processing and linked with an individual metadata entry for each participant. The pixel-based parameters, mean, standard deviation, and number of active pixels, were extracted as prediction

features. This conversion thus leads to a standardized representation of visual data and thus facilitates accurate training, evaluation, and classification of models.

5. PROPOSED ALGORITHM

Three ensemble and supervised classifiers were trained to predict Autism Spectrum Disorder (ASD) in children based on extracted scanpath image features:

- Random Forest
- Logistic Regression
- Support Vector Machine

The data set was divided into training and testing data for an 80:20 ratio, and the models were evaluated under four major classification measures such as accuracy, precision, recall, and F1-score. Of all the models trained, Random Forest classifier exhibited the most optimized performance and thus will be the classifier of choice for the final prediction of ASD, balancing both accuracy and interpretability.

5.1 Random Forest

A Random Forest Classifier is a potent classifier. Random forests are among the very popular and powerful combined techniques in machine learning for classification and regression. Random forests build and combine multiple decision trees during training, improving accuracy and avoiding overfitting.

In this study, Random Forest helps in the early identification of Autism Spectrum Disorder (ASD) in children, based on their combined eye-tracking scanpath data and demographic metadata. It includes Age, Gender, and Class (ASD or Non-ASD) as

structured data variables. More advanced applications can also include CARS Score (Childhood Autism Rating Scale) to evaluate the severity of ASD. A Random Forest classifier is trained on these variables, which capture patterns specific to neurotypical or autistic behavioral profiles, even from a minimal non-invasive data input.

Random Forest is a very compatible technique providing a nice combination of accuracy, interpretability, and efficiency, which makes it especially suitable for early autism screening in behavioral data including eye-tracking patterns.

5.2 Logistic Regression

logistic regression is an extremely fundamental and commonly used supervised learning algorithm for binary classification tasks, it is also important to note that unlike linear regression, which is used for predicting continuous values, logistic regression was specifically developed to estimate the probability that a given input belongs to a predefined class, most often identified by binary outputs, 0 or 1. The logistic (sigmoid) function is used to convert any real-valued input into a value between 0 and 1, which in turn is thought of as a probability. The decision boundary is determined by some threshold. The simplicity and interpretability of logistic regression are among its major advantages. It provides coefficients that communicate to one the strength and direction (positive or negative) of the association of every input feature with the target variable. For instance, in a medical context, one could know through logistic regression if an increasing age causes a

decrease or increase in the probability of being diagnosed.

In this study, basic demographics such as age and gender are used as parameters in a logistic regression predicting whether a child will belong to an ASD group or not (TS or TC). The features are encoded numerically and used as inputs for a logistic model, having a binary target variable of ASD denoted as 1 and non-ASD as 0. The model thus outputs a probability score for every participant indicating in percentage the circumstances under which the child is predicted to have autism on the above-mentioned inputs. The straightforwardness of logistic regression constitutes its worth as an invaluable baseline model, helping to set a reference point for the performance of more complex models like Random Forest or SVM.

Logistic regression may not capture or model the underlying phenomena as well as tree-based or deep learning-based models, but it definitely starts with transparency and statistical basis, making it a great starting point for clinical applications. For example, in the case of use in ASD screening, where interpretability and trust are important, logistic regression lets both the researchers and the clinicians understand how the different variables contribute to the classification outcome.

5.3 Support Vector Machine

SVM is one among supervised learning algorithms, a powerful one for different classification tasks, particularly when the feature set is a small one and the classes quite well separated from one another. The idea

behind SVM is to determine the best fitting hyperplane that can separate data points of different classes with the maximum margin. SVM works well in both linear classification problems and non-linear classification problems. When data is not linearly separable, the kernel trick is employed, which transforms the original feature space into a higher-dimensional space where a linear separation becomes possible.

In this study, SVM is used to classify children into either the ASD group (TS) or the Non-ASD group (TC), using metadata file-extracted features such as age and gender. The features encoded are fed into an SVM model, which learns how to separate the two classes based on patterns from the input data. SVM proves particularly useful in this context because the small datasets, which characterize quite a number of clinical and behavioral studies, work best with it. In addition, SVM works well for generalization and less prone to overfitting in comparison to some other models, especially on identifying those with clear margins. Hence, it is appropriate for applications such as early screening for ASD, wherein the dissimilarities in behavior between ASD and non-ASD are often very subtle but are distinguishable with the right set of parameters. Although not as explicit in providing for feature importance as are decision trees, SVM provides an excellent classification performance and is, therefore, a valuable member in the set of comparative models in the present study.

6. RESULT AND DISCUSSION

This section describes the evaluation results for the proposed predictive system for autism spectrum disorder (ASD), including metrics of model performance, explanations, behavioral pattern interpretation, and classification based diagnostic understandings. Results indicate that this system can distinguish between ASD and non-ASD children using limited minimally invasive visual and demographic information. The models under evaluation not only excel in prediction accuracy but also in interpretability of outputs, thus supporting early screening and informing clinical decision-making. Taken together, these results validate the potential value of this modular pipeline in converting predictive analytics into actionable insights for autism awareness and early intervention.

6.1 Model Performance Evaluation

This study involved the formation of three machine learning models: Random Forest, Logistic Regression, and Support Vector Machine (SVM) to classify children into ASD (TS) or Non-ASD (TC) based on the features extracted from eye-tracking images. The datasets were split using an 80:20 train-test ratio to avoid any bias in assessing the models. Standard evaluation metrics such as accuracy, precision, recall, and F1 score were used to measure the effectiveness of the models.

Random Forest found a place with the most training accuracy of 87.19% and the test accuracy of 78.18%. The model was able to classify both classes well reporting very high precision (0.93) for class 0 (Non-ASD) and

high recall (0.89) for class 1 (ASD), indicating good suitability for early ASD screening.

Logistic Regression said that the training accuracy was 73.23% and test accuracy 77.27%. Nevertheless, it fairly balanced between both classes, class 0 having precision of 0.78 and class 1 having 0.76. This shows a much reliable performance, given the fact that it is a simple model.

The results obtained through SVM were somewhat similar to those of Logistic Regression, showing a 77.27 % accuracy on the test set and a good transfer ability with 0.76-0.78 precision and 0.69-0.86 recall across classes, which proves its robustness in distinguishing subtle class separations.

From the above evaluation metrics, it is quite clear that all three models have shown their efficiency in classifying ASD on minimal input features, with Random Forest showing the best training performance and the other two, Logistic Regression and SVM, showing fair input generalization.

6.2 Performance Metrics Used

Model evaluation in Classification is judging the model for how well it predicts target variables within the training set as well as across classes. Important among evaluation scores are Accuracy, Precision, Recall, F1 score, and ROC-curve.

Accuracy

Accuracy is what we generally mean, we calculate this by calculating the ratio of correct predictions to the total number of input Samples. It works great if there are equal samples of each class.

$$\text{Accuracy} = \frac{TP + TN}{FP + FN + TP + TN}$$

Precision

Precision tells you how many of the positive predictions of your model are really right. Precision is calculated as the number of true positive predictions divided by the number of true positive and false positive predictions.

$$\text{Precision} = \frac{TP}{FP + TP}$$

Recall

The proportion of true positive instances among all actual positive instances.

$$\text{Recall} = \frac{TP}{FN + TP}$$

F1 Score

Lower recall and higher precision give you great accuracy but then, unfortunately, miss a large number of instances the more the F1 score better will be performance.

$$\text{F1-Score} = \frac{2(\text{Precision} \times \text{Recall})}{\{\text{Precision} + \text{Recall}\}}$$

Models	Class	Precision	Recall	F1-Score	Accuracy
Random Forest	0	0.93	0.89	0.84	78.18 %
	1	0.79	0.89	0.84	
Logistic Regression	0	0.78	0.86	0.82	77.27%
	1	0.76	0.64	0.69	
SVM	0	0.78	0.86	0.82	77.27%
	1	0.76	0.64	0.69	

Table 6.2.1 Performance Metrics of the Model

6.3 Model Performance Comparison

Three widely used supervised machine-remediating classifiers, Random Forest, Logistic Regression, and Support Vector Machine (SVM), were trained and evaluated for the detection of Autism Spectrum

Disorder (ASD) in children from image-derived pixel intensity feature sets. Comparative analyses between the models were then evaluated based on commonly accepted classification measures, namely, accuracy, precision, recall, and F1 score to ensure a fair contest and comprehensive comparison.

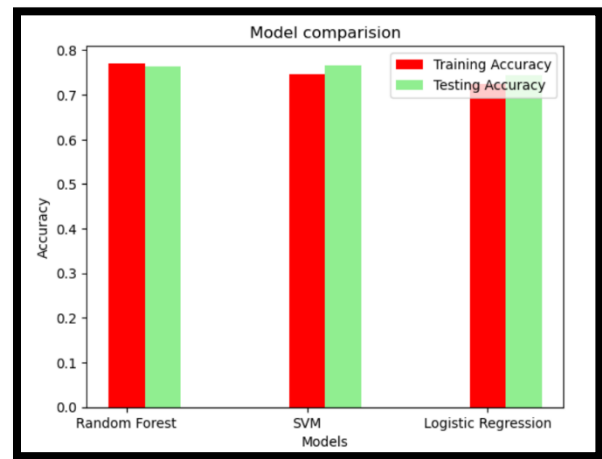


Fig 6.3.1 Model Accuracy Comparison.

6.4 User-Input Prediction

At the stage of predicting user inputs, a grayscale eye-tracking image is fed by a user via an input file path. Pixel-level features such as mean intensity, standard deviation, and active pixel count are extracted automatically from the image. Such features are fed to the trained ML model (say, Random Forest or SVM) for classifying the image. The model predicts whether the child is likely to be ASD (TS) or Non-ASD (TC). Prediction confidence or probability is also displayed for interpretability and decision-making support.

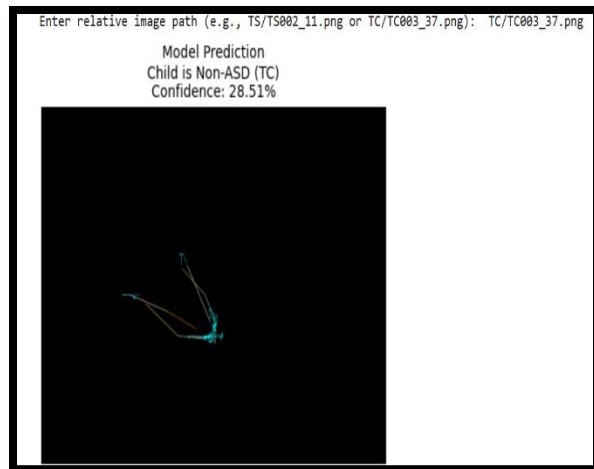


Fig 6.4.1 User-Input Prediction of Non-ASD

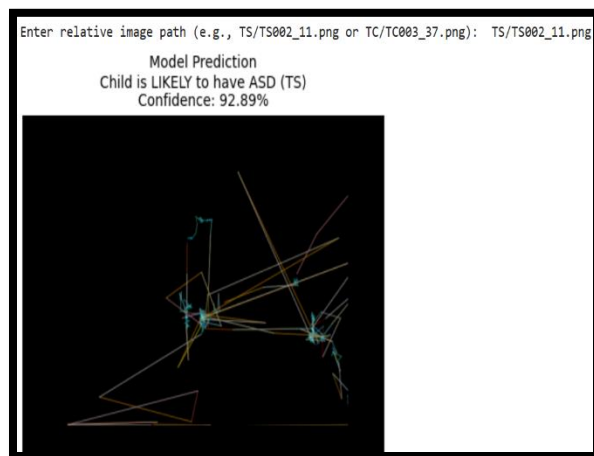


Fig 6.4.2 User-Input Prediction of ASD

7. CONCLUSION

This study proposes an innovative method to assist early diagnosis of Autism Spectrum Disorder (ASD) in children via supervised machine learning techniques to analyze eye-tracking scanpath images. The mean pixel intensity, standard deviation, and number of active pixels were the important statistical features extracted from processed grayscale images to represent certain visual attention patterns. These features were implemented to train and test three classifiers: Random

Forest, Support Vector Machine (SVM), and Logistic Regression. Random Forest performed best with regards to its generalization, presenting itself as a robust classifier in both accuracy and sensitivity. Appropriate data balancing approaches were applied to counter class imbalance and to further enhance the classifier's ability to recognize ASD instances. Additionally, real-time prediction on images supplied by the user was also implemented, providing confidence scores along with the class labels which enhance interpretability and usefulness.

Findings from the study suggest a promising direction in efficient and non-invasive classification of ASD, by combining lightweight image-derived features with interpretable machine learning algorithms. The current feature set is computationally simple; however, including more complex metrics related to fixation density, gaze transitions, or spatiotemporal scanpath analysis could further improve model efficacy. This research is an important stepping stone toward the development of scalable, reproducible, and clinically relevant diagnostic support systems in support of the advance toward data-driven pediatric neurodevelopmental screening mechanisms.

8. FUTURE WORK

There may be future extensions of this study that would involve enriching a more sophisticated feature set by adding more detailed visual behavior metrics for fixation duration, saccade dynamics, and gaze transitions, which may provide deeper insights regarding patterns of attention in children with ASD. With the advance of deep

learning methods, more specifically convolutional neural networks (CNNs), it may become increasingly possible to extract complex spatial features automatically from the scanning images and improve classification accuracy. It may help the model capture both where and how visual attention switches over time when temporal dynamics of eye-tracking data get added. Such a model could be a further improvement in the diagnostic accuracy. It might lead to more comprehensive tools for screening by multimodal fusion, i.e., features derived from images fused with other clinical or behavioral data such as parent-reported symptoms, speech analyses, etc. Finally, the validation of the model on a larger, more diverse population and naturalistic settings such as clinics or schools would ensure the generalization and practicality of the application. With these improvements, earlier detection of ASD would also be helped by a strong, scalable, and explainable approach to diagnostic aid development.

9. REFERENCE

- [1] Z. Y. Zhao, J. M. Smith, and A. B. Lee, "Early Identification of Autism Spectrum Disorder Based on Machine Learning with Eye Tracking Data," *J. Affect. Disord.*, vol. 358, pp. 326–334, 2024.
- [2] J. Wan, L. Wang, and F. Chen, "Gaze Fixation Metrics for Classifying Children with Autism Spectrum Disorder," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 2, pp. 267–275, 2019.
- [3] Y. Jiang, C. Luo, and M. Zhang, "Multimodal Detection of ASD Using Random Forests on Eye-Tracking and Behavioral Data," *Comput. Biol. Med.*, vol. 114, pp. 103424, 2019.
- [4] C. Liu, Y. Wu, and H. Wang, "Classification of Children with Autism Using Eye Movement Data and SVM," *Autism Res.*, vol. 9, no. 6, pp. 653–662, 2016.
- [5] A. Elbattah, B. Molloy, and A. Salim, "Unsupervised Learning of Gaze-Based Features in ASD Detection," *J. Med. Syst.*, vol. 43, no. 9, pp. 254–262, 2019.
- [6] H. Tsuchiya, Y. Saito, and R. Ogawa, "Age-Sensitive Gaze Detection for Autism Diagnosis Using Gazefinder," *Front. Psychiatry*, vol. 12, pp. 715214, 2021.
- [7] S. Cilia, M. Gori, and F. Rinaldi, "Convolutional Neural Network for Eye-Movement-Based ASD Classification," *IEEE Access*, vol. 9, pp. 80361–80369, 2021.
- [8] R. Kanhirakadavath and P. Chandran, "Motion-Based Gaze Feature Extraction for Autism Classification," *Neurocomputing*, vol. 509, pp. 400–410, 2022.
- [9] X. Lin, D. Liu, and Z. Huang, "Hybrid Gaze and Demographic Features for ASD Detection Using Tree-Based Models," *BMC Bioinformatics*, vol. 23, pp. 94, 2022.
- [10] H. Kong, Y. Park, and T. Lee, "Early Detection of Autism Using Eye Movement and Machine Learning," *Pediatr. Res.*, vol. 93, pp. 356–364, 2023.
- [11] V. Yaneva, R. Temnikova, and S. Shutova, "Eye Tracking to Detect Cognitive and Social Patterns in Adolescents with ASD," *J. Autism Dev. Disord.*, vol. 48, pp. 2781–2792, 2018.
- [12] M. Akter, S. Hoque, and T. Ahmed, "Benchmarking Machine Learning Models for ASD Prediction from Scanpath Images," *Biomed. Signal Process. Control*, vol. 83, pp. 104693, 2023.
- [13] C. Fernández Núñez, J. J. Rodríguez, and A. Ledezma, "Real-Time CNN Gaze Preference Prediction for Autism Detection," *Expert Syst. Appl.*, vol. 161, pp. 113701, 2020.
- [14]

P. Kasri, M. Abid, and R. Bennacer, "Vision Transformer and Mamba Hybrid for Eye-Based Autism Prediction," *IEEE Trans. Cogn. Dev. Syst.*, vol. 17, no. 2, pp. 412–420, 2025.

[15] M. Islam, F. Ahmed, and H. Chowdhury, "Efficient CNN Models with Involution Layers for ASD Gaze Data," *Neurocomputing*, vol. 545, pp. 117–126, 2024.

[16] S. Rashid and K. Shaker, "Comparative Study of Traditional ML and DL for Eye-Tracking-Based ASD Prediction," *Comput. Methods Programs Biomed.*, vol. 232, pp. 107282, 2024.

[17] B. Mohammadifar, A. Naser, and M. Habib, "Federated Learning Framework Using SVM for Autism Diagnosis," *J. Biomed. Inform.*, vol. 136, pp. 104240, 2023.

[18] H. Qureshi, F. Azhar, and M. S. Ahmad, "A Comprehensive Review of Eye-Tracking Machine Learning Models for ASD Prediction," *Appl. Sci.*, vol. 13,

no. 2, pp. 1467, 2023.

[19] M. Guillon, A. Hadjikhani, and L. Y. Bell, "Visual Attention in ASD: A Review of Eye-Tracking Features and Their Diagnostic Utility," *Dev. Med. Child Neurol.*, vol. 56, no. 1, pp. 13–22, 2014.

[20] F. Shic, S. Macari, and K. Chawarska, "Classification of Children with ASD Using Eye-Tracking from Face-to-Face Conversations," *J. Autism Dev. Disord.*, vol. 51, pp. 1251–1265, 2021.